# RYOTARO NIWA

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## RESEARCH INTEREST

**Quantum Information Theory**
Quantum algorithms, Quantum channel learning, Quantum error-correction

## EDUCATION

**PhD** *April 2026-Present*
University of Tokyo, Supervisor: Prof.Mio Murao

**MSc** *April 2024-March 2026*
University of Tokyo, Supervisor: Prof.Mio Murao
Thesis: "Quantum algorithms for learning the spectrum of quantum channels"

**BSc** *April 2020-March 2024*
University of Tokyo, Major: Physics

## PUBLICATION & PREPRINTS

* denotes first author

[1] **R.Niwa***, S.Yoshida and M.Murao, *Asymptotically optimal purification of noisy unitary channels in any dimension*, arXiv:2608.26061 (2026)

[2] **R.Niwa***, S.Yoshida, K.Ono, T.Utsumi, Z.Li, Y.Yang, R.Takagi and M.Murao, *Scaling-optimal purification of noisy qubit unitary channels*, arXiv:2606.12394 (2026)

[3] S.Yoshida, **R.Niwa**, and M.Murao, *Random dilation superchannel*, arXiv:2512.21260 (2025)

[4] **R.Niwa***, Z.M.Rossi, P.Taranto, and M.Murao, *Singular value transformation for unknown quantum channels*, arXiv:2506.24112 (2025)

[5] **R.Niwa*** and J.Y.Lee, *Coherent information for Calderbank-Shor-Steane codes under decoherence*, PhysRevA.111.032402 (2025)

## RESEARCH TALK & SEMINARS

- "Scaling-optimal purification of noisy qubit unitary channels", Seminar talk at the Technical University of Munich, June 2026
- "Singular value transformation for unknown quantum channels", Seminar talk at the University of Edinburgh, February 2026

## PRESENTATION

**Contributed Posters**

- 26th Asian Quantum Information Science Conference (AQIS 2026), Daejong, August 2026
- 7th Seefeld Workshop on Quantum Information, Seefeld, June 2026
- 29th Annual Quantum Information Processing Conference 2026 (QIP 2026), Riga, January 2026
- 25th Asian Quantum Information Science Conference (AQIS 2025), Hong Kong, August 2025
- Quantum Innovation 2025, Osaka, July 2025

- From quantum materials to quantum information (QMQI 2024), Okinawa, November 2024
- Quantum Innovation 2024, Tokyo, October 2024

## AWARDS AND FELLOWSHIPS

**Excellent Research Assistant** *2024-2026*
Q-leap Quantum Information Processing

**Scholarship Award** *September 2023*
Scholarship for the Study and Visit Abroad Program (SVAP)

## TEACHING EXPERIENCES

**Quantum Mechanics II** *Spring 2024*
Teaching Assistant (TA)

**Exercise in physics I** *Fall 2024*
Teaching Assistant (TA)

## COMMUNITY SERVICE

Reviewer for Quantum, IEEE J-STEP.